

25. NEURAL TUBE CLOSURE

DURATION : 03:10

STUDY SHEET

COURSE SUMMARY

This session explores in detail the closure of the neural tube, a crucial moment in embryonic development that begins with the formation of an elongated structure from the epiblast. This process involves the fascinating interaction between the hypoblast and the ectoderm, leading to the emergence of the notochord and its transformation into a chordal rod. Students will learn how this dynamic influences vertebral formation and integrates essential information for embryonic development. By understanding these interactions, therapists can better grasp the embryological foundations related to their practices.

NEURAL TUBE CLOSURE

Neural tube closure is a crucial process in embryonic development. This tube, by its nature, is hollow and forms from the epiblast.

Initially, the tissue deposits on the epiblast, then closes to form an elongated structure. Beneath this structure lies the **hypoblast**, which can be considered the "mouths" of the **endoderm**.

As one moves towards the **endocoel**, the shape of the tube becomes more defined. This growth process leads to a flattening on the hypoblast, followed by a regaining of height and a restructuring into a cord. It is essential to visualise this formation as a tube.

Observing a cross-section, one can see the inside of this structure after removing the epiblast. At this stage, the **ectoderm** and **endocord** can be placed above, while the structure flattens against the hypoblast.

At some point, this structure completely inserts itself between the other tissues and becomes a cord without an internal lumen. It is at this stage that it acquires **ecto-mesodermal** characteristics, but this primarily concerns the **notochord**, and not the **mesoblast**.

The notochord, at this stage, is considered a chordal tube, subsequently evolving into a chordal cord. Although it is epiblastic, during its closure, it integrates hypoblastic information, but remains essentially epiblastic.

This dynamic is important: the notochord is epiblastic due to its development. When it becomes chordal, it retains hypoblastic information.

This information is crucial, as it constitutes the **neural tube**, around which the vertebra is outlined. One can already observe the vertebra forming around the notochord, which then leaves a vestige of its existence.